

NAGHAM NASSER KAMAL

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EDUCATION

Bachelor of Communication and Computer Engineering, Benha University

Oct 2019 - Jul 2024

- **GPA:** 3.6/4.0
- **Relevant Coursework:** Design of Electronic Circuit, Signal and Systems, Communication Systems, Information Systems, Data Structures and Algorithms, Computer Organization, Advanced Computer Architecture, Object Oriented Programming, Mathematics, Embedded Systems.

SKILLS

Programming Languages	Python, C, C++, MATLAB, VHDL
Technical Skills	English: Proficient , Arabic: Native, Communication Systems, Encryption Algorithms, Algorithms and Data Structures, Artificial Intelligence, OOP, Machine Learning, Statistics, Mathematics.
Soft Skills	teamwork, emotional intelligence, conflict resolution,
Certifications	C and Embedded C , AI Machine Learning Specialization , AI Deep Learning Specialization .

EXPERIENCE

C and Embedded C Programming, NTI - National Telecommunication Institute

Aug 2023

- Mastered the fundamentals of the C programming language, including essential data structures such as stacks, queues, and linked lists, along with key algorithms for searching and sorting.
- Explored the fundamentals of interfacing with microcontrollers.
- Developed and implemented a School Management System as the final project, using linked lists to manage data efficiently. This project was executed independently.

Deep Learning Specialization

Aug,2022 - Sep,2024

- Acquired expertise in understanding and implementing neural networks, leveraging Python, NumPy, and TensorFlow for model training and deployment.
- Innovated a custom optimization algorithm that combined gradient descent with adaptive learning rates.
- Utilized convolution neural networks (CNNs) for computer vision applications, including image recognition and object detection.

PROJECTS

- **Modulation Techniques in MATLAB:** Developed a project on modulation techniques including Amplitude Modulation (AM), and Frequency Modulation (FM) using MATLAB. It involves analyzing for these modulation schemes, studying their spectral properties, and enhancing the audio by 80%.
- **Data Encryption Standard (DES):** Implemented the DES cryptographic algorithm on the 8051 microcontroller, optimizing it in C for a 30% improvement in the memory. The project involved handling 64-bit data blocks to enhance the performance for security-critical applications.
- **Car Logo Detection under Complex Lightning Conditions:** Utilize YOLOv7 and YOLOv8 models for tracking the car logos in various lighting conditions, with the help of different augmentation techniques as EnlightnGAN, to enhance the efficiency of the detection. The usage of our custom dataset, and YOLOV8 that specifies in detecting small objects enhance the map to reach 87.3%.
- **VHDL Pipelined MIPS Processor** :Engineered a 24-bit pipelined MIPS processor using VHDL, which includes a sophisticated control unit for comprehensive hazard detection. Implemented full data forwarding mechanisms to efficiently handle data hazards and enhance pipeline performance. This design ensures seamless operation by minimizing stalls and optimizing data flow through the processor.

GRADUATION PROJECT

- **Objective:** Developed from scratch fully self driving car using software simulation and hardware.
- **Tools:** Arm microcontroller, AVR microcontroller, CAN Module, C language, Flutter, Python, BA-HPC, CARLA Simulator, NVIDIA Jetson nano, and Rssbery pi camera V2.
- **Description:** Our project consists of four stages aimed at developing a comprehensive autonomous car system
 - **Manual Control:** Implemented a manual control system for the car using a mobile application, enabling direct user interaction.
 - **Anti-Collision System:** Developed a safety mechanism to validate user actions before car movement, maintaining the vehicle within a safe zone. This system reduced potentially hazardous actions by 90%.
 - **Lane Keeping:** Utilized a reinforcement learning model to ensure correct lane adherence based on real-time images captured by an onboard camera. It also employs a PID controller combined with a reinforcement learning model to stabilize the car, achieving a model efficiency exceeding 92%.
 - **Self-Driving Capabilities:** Enhanced the lane-keeping system to enable autonomous driving, allowing the vehicle to change lanes and navigate in various directions using advanced AI and reinforcement learning models.
- **My Contributions:**
 - Worked with ARM architecture to interface with multiple sensors, including GPS and GSM, for accurate environmental awareness.
 - Developed a detection model using YOLOv8, fine-tuning it with a custom dataset to achieve a mean average precision (mAP) of 85%.
 - Created a segmentation model with the Hugging Face library, achieving an accuracy of 92%.
 - Set up the development environment on NVIDIA Jetson Nano, utilizing CUDA for GPU processing, reducing detection time by 100x compared to CPU processing.
 - Integrated all components and models to complete the final autonomous driving system.